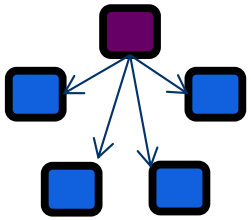
CompSci 390-03-26S

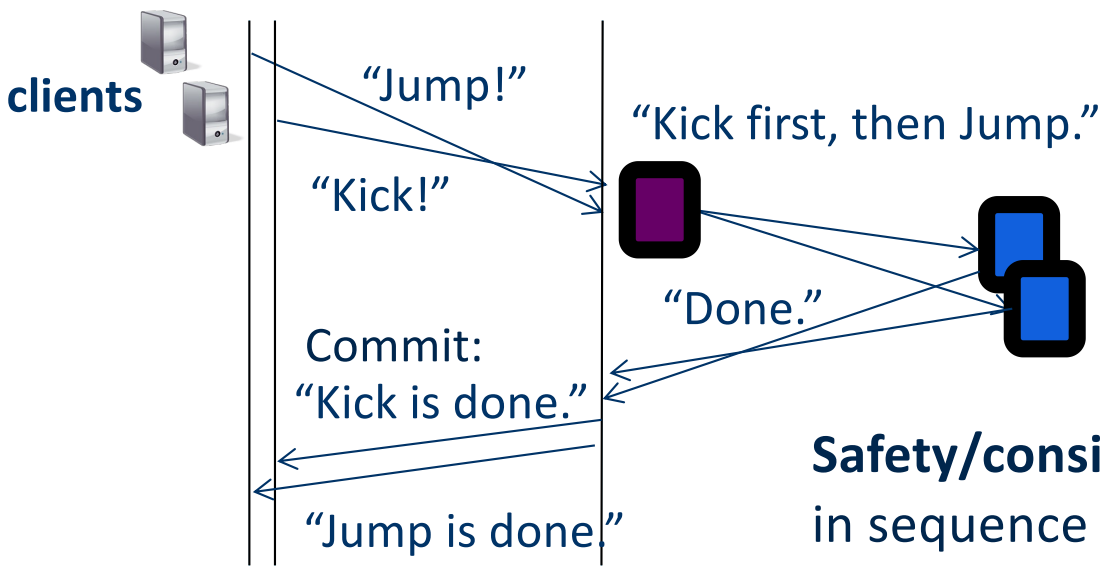
Distributed and Networked Systems

State Machine Replication and Raft Consensus



Primary/backup replication: the story so far

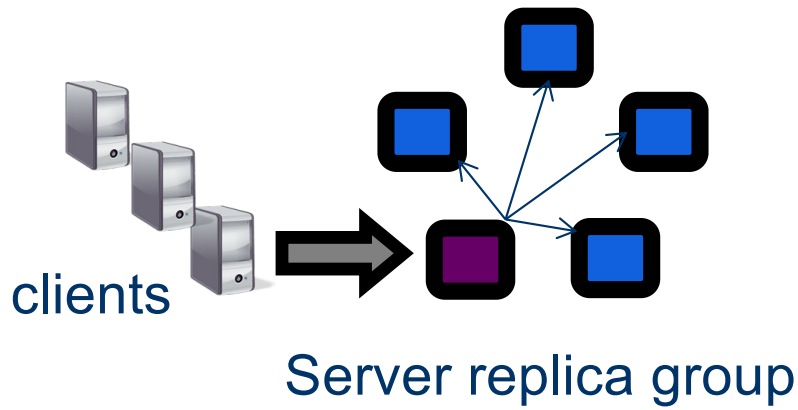
- Clients send requests to one server (**primary** or **leader**).
- Primary sequences the requests and pushes them to **backup** replicas (**followers**) before responding to clients.



Safety/consistency: ops commit in sequence and committed ops never revert.

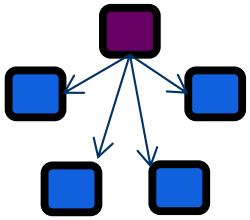
Replicated service: context/assumptions

- Benefits/purpose of replication?
- Assumptions?
  - About the application service?
  - About client interactions with servers?
  - About server group and server failures?
  - About the network?
- Are these assumptions "right"?
- Relevant in practice?
- **What makes it hard?**



Consensus

- **Consensus** is a class of problems and distributed protocols involving agreement among a set of participating processes/servers.
- **Example**: State machine replication (SMR/RMS) in which servers in a replica group agree on a sequence of operations to apply to replica state.
  - Equivalent to a **replicated log** of operations
  - Example 1: **cloud** state services—Zooky and Spiffle
  - Example 2: **blockchain—**"distributed ledger"

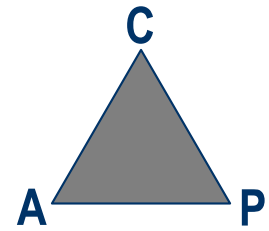


- Various failure models and impossibility results
  - **Cloud**: FLP and CAP for fail-stop failures in asynchronous networks
  - **Blockchain**: Lamport/Pease for Byzantine failures (conspiracies)

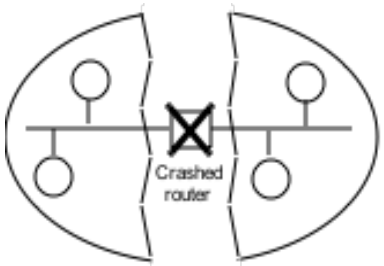
Impossibility of consensus: CAP

Consensus can be safe or live but not both.

- In an asynchronous network with failures and **partitions**
- Safe (consistent) consensus may occur recognizably, but liveness (availability) is not guaranteed.
- Fischer-Lynch-Patterson (1985)
- Proof intuition: a "failed" server might just be "slow"—and may rise from the dead at an inconvenient time.



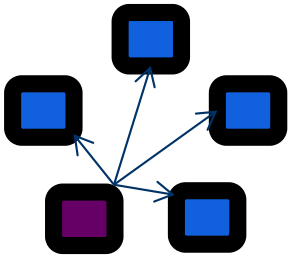
Choose two.



Network partition

FLP popularized as Brewer's Conjecture or CAP trilemma. **No you can't have it all**: **C**onsistency + **A**vailability in presence of **P**artitions.

Consensus replication: Five assumptions



1. **Deterministic** application service (whatever it is).
2. **Known configuration**: fixed group of N=2f+1 replicas.
   - Can change it through the protocol, but it is out of scope for us.
3. **Ready quorum**. Available only if majority (f+1) are up and connected. Else CAP says: have another round.
4. **Fail-stop faults**: a failed replica is silent. Replicas run the same program and are faithful to it. Net may delay.
5. **Smart clients**. On fail-over, clients switch and resync to the new leader. No lost/duplicate requests or responses.

Marked slides incorporate content from the authors of the Raft paper.

In Search of an Understandable Consensus Algorithm

Diego Ongaro and John Ousterhout, *Stanford University*

https://www.usenix.org/conference/atc14/technical-sessions/presentation/ongaro

This paper is included in the Proceedings of USENIX ATC '14: 2014 USENIX Annual Technical Conference.

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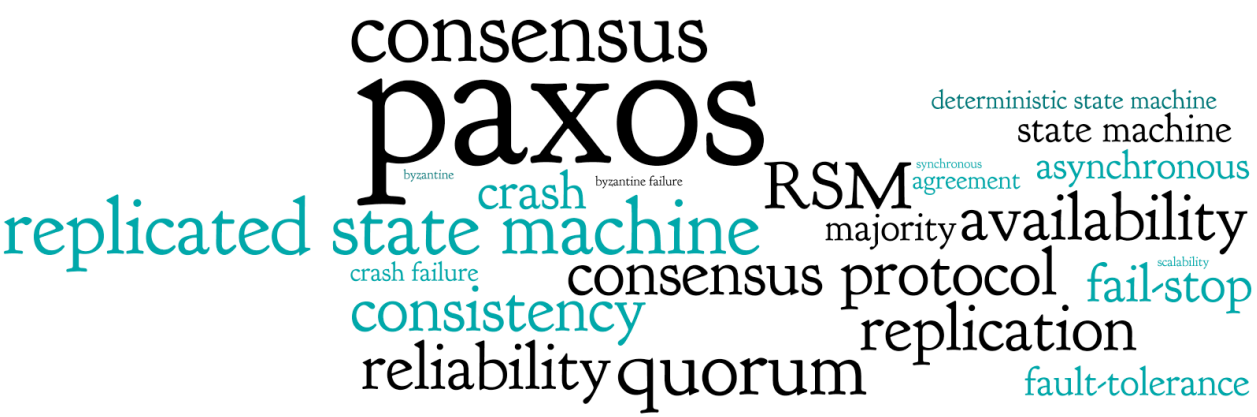
Understandable?

PAXOS

Made Moderately Complex

http://paxos.systems "PMMC"
"This website explains the infamously difficult to understand Paxos consensus protocol using easy to understand invariants and code."



paxos inside

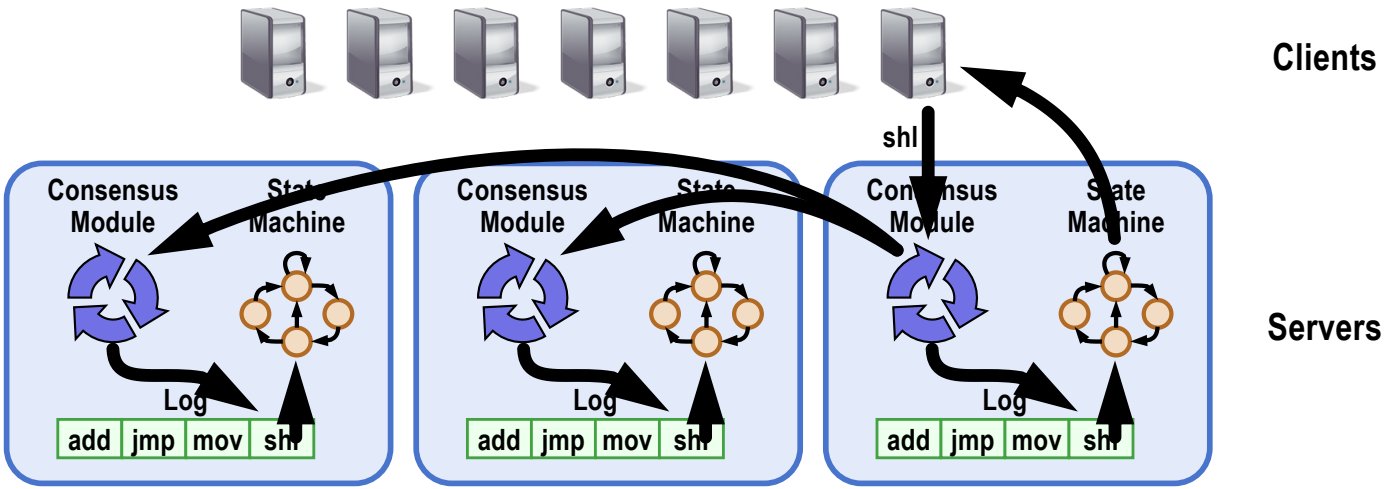
Chapter 7
From Viewstamped Replication to Byzantine Fault Tolerance

Barbara Liskov

Recommended! Prof. Liskov's 2010 book chapter summarizing her landmark work with Brian Oki on VR (1988), contemporaneous with Paxos, and Practical Byzantine Fault Tolerance (PBFT, 1999), a foundation for modern permissioned blockchains.

Raft is similar to VR, but the terms differ. 😄 Also certain details of leader selection and log replication.

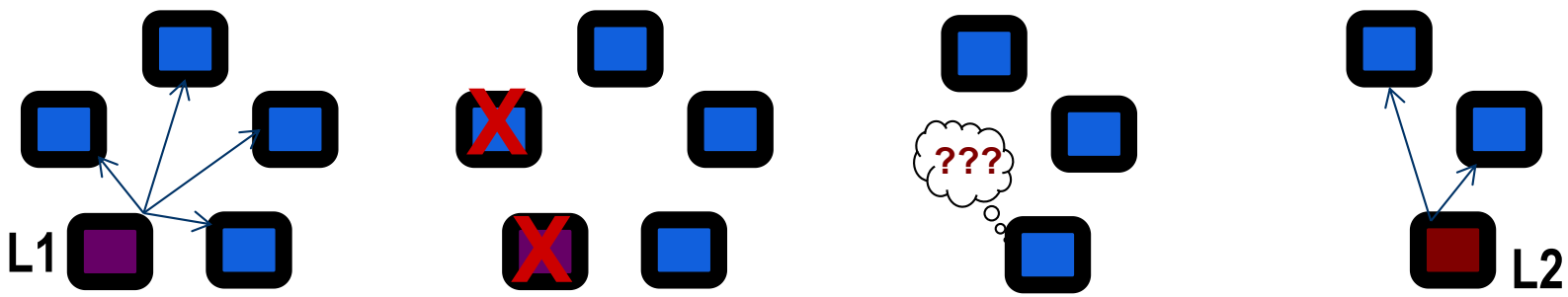
Goal: Replicated Log



- Replicated log => replicated state machine
  - All servers execute same commands in same order
- Consensus module ensures proper log replication
- System makes progress as long as any majority of servers are up
- Failure model: fail-stop (not Byzantine), delayed/lost messages

Stable leader: Handling leader failure

- Leader sends a message to followers at regular intervals, even if there are no new op requests.
  - Called **heartbeat** or **ping** or **keepalive**.
- Followers set timers. If timer fires with no heartbeat, abandon the leader and initiate **leader election**.



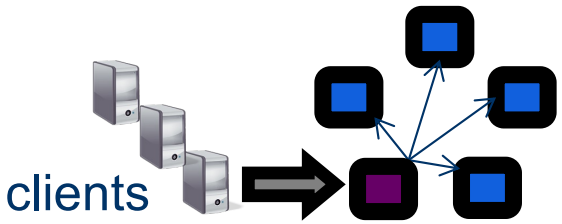
A happy group of 5 replicas.

Suppose leader and another replica fail.

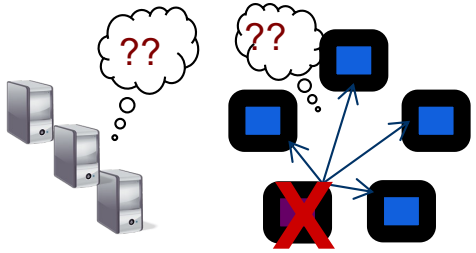
Time out!!! Where is L1?

The survivors elect L2 as leader. **Keep going.**

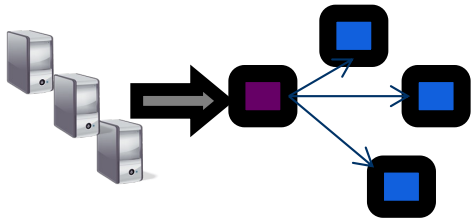
Failover for a replicated service



1. A **leader** (primary) server receives requests, sequences them, and passes them to one or more **followers**. All execute the requests in order, but only the leader responds to clients.



2. If the leader fails, others detect the failure by **timeout**. A **leader s/election** designates a follower as the new leader. It has "the same" state as the failed leader and is "indistinguishable".

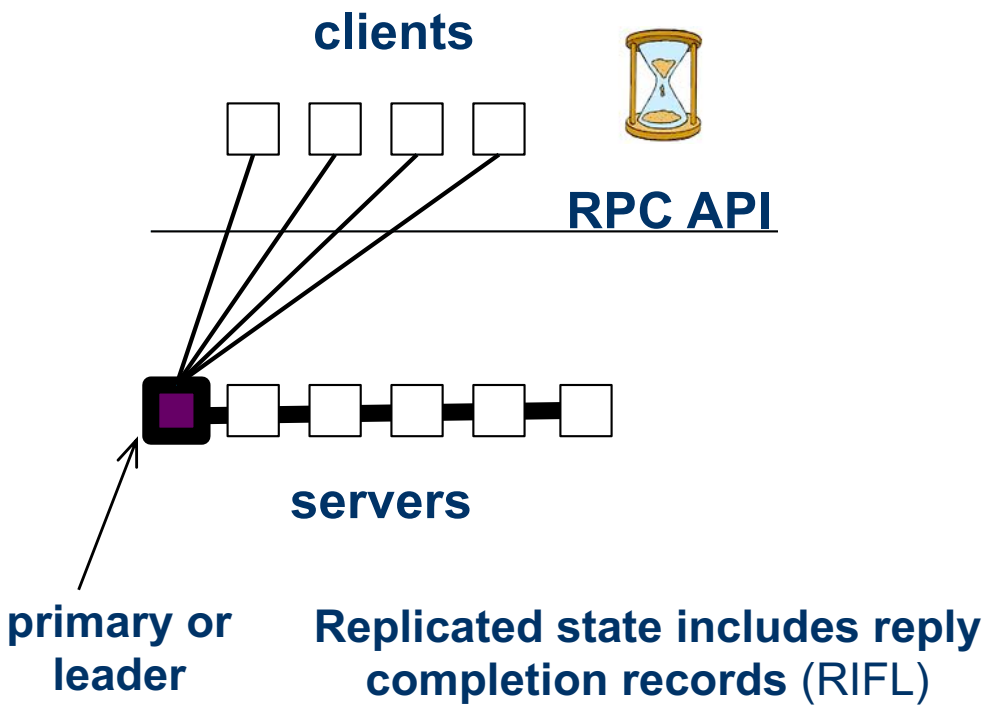


3. Each client finds and connects to the new leader, and **retransmits** any pending RPCs. The leader **filters duplicates**, and continues from where the old leader left off (**failover**).

Notes

Reliable RPC: recap #1

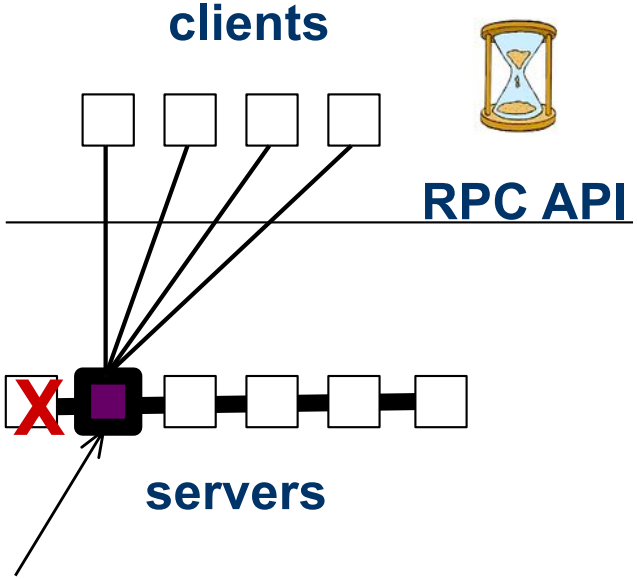
- Clients identify primary server.
- Clients direct RPCs to primary.
- Clients tag each call with a sessionID and callID (seq#).
- Server places call tag in reply to match calls/replies.
- Client and server retransmit all messages until they are acked (or the client fails).
- Client and server suppress duplicate calls/replies.



Notes

Reliable RPC: recap #2

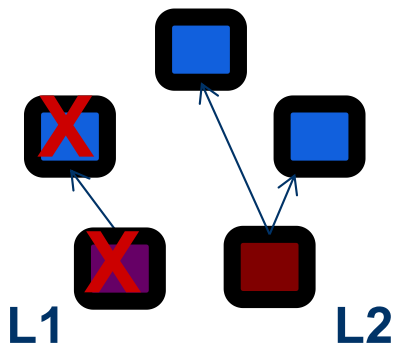
- If primary fails, service **fails over** to a backup server.
- The backup becomes the new primary server.
- **Reply completions and other state transfer to new primary.**
- Clients **retarget** to new primary.
- RPC session continues the stream of calls/replies.
- Transparent to application
- Client failure/recovery is made visible via session leasses for AMO.



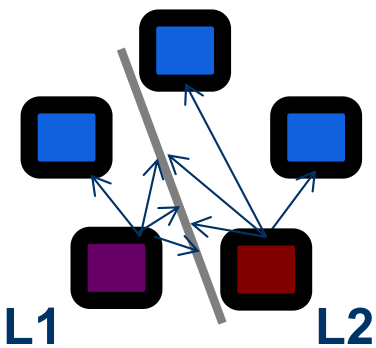
primary or leader

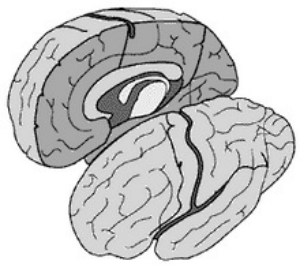
Replicated state includes reply completion records (RIFL)

The Curse of CAP



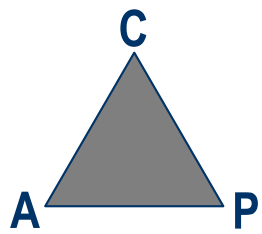
If servers fail, the survivors must take over to serve requests. **Required for Availability (A).**





But what if a **P**artition?
Split brain!
Not Consistent (C)

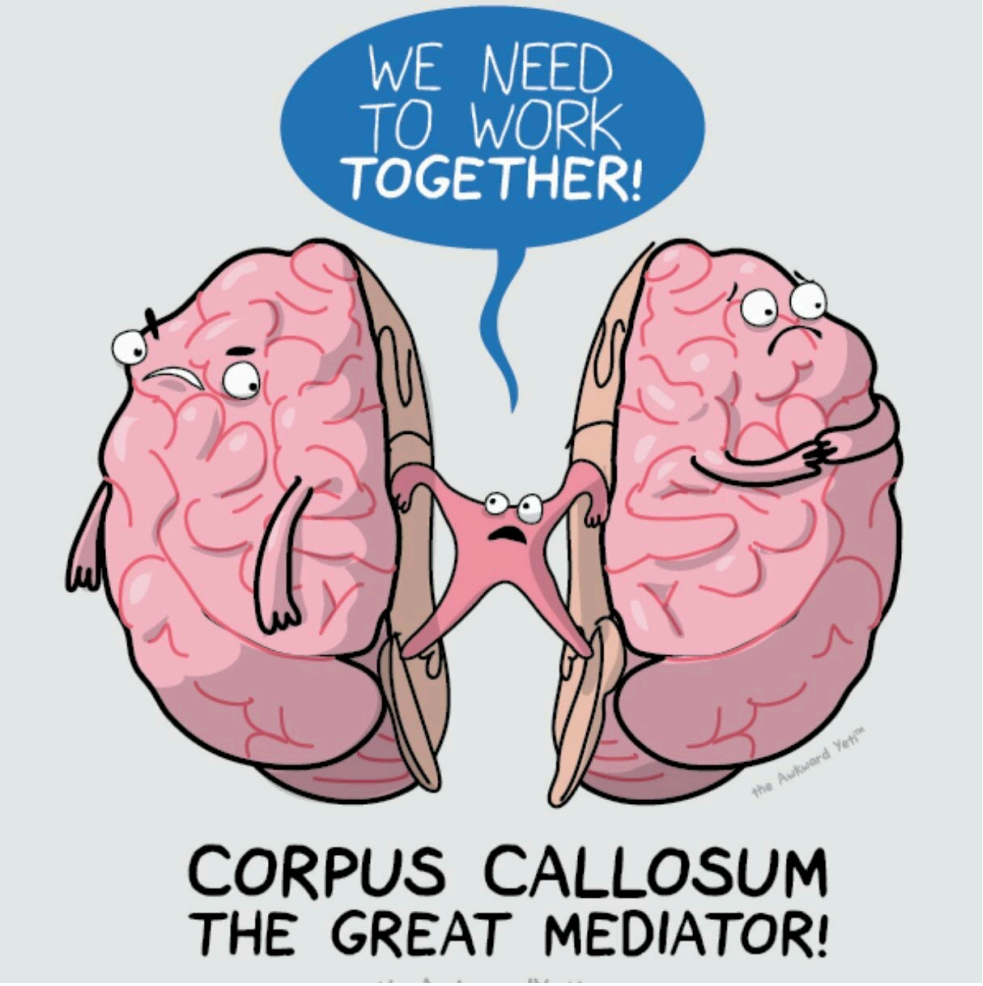
Who can tell the difference?



Choose two.

"Split brain"

- **Split brain** is a CS term of art.
- Inspired by real brains:
  - Two cerebral "hemispheres"
  - Operating semi-independently
  - Narrow communication channel
  - Damage to corpus callosum has debilitating impacts.
  - "Left hand doesn't know what the right hand is doing."



https://me.me/i/we-need-to-work-together-corpus-callosum-the-great-mediator-398fd3c0f98643b99c2fbf980a1c237b

One option: Keep going and fix it later C→AP

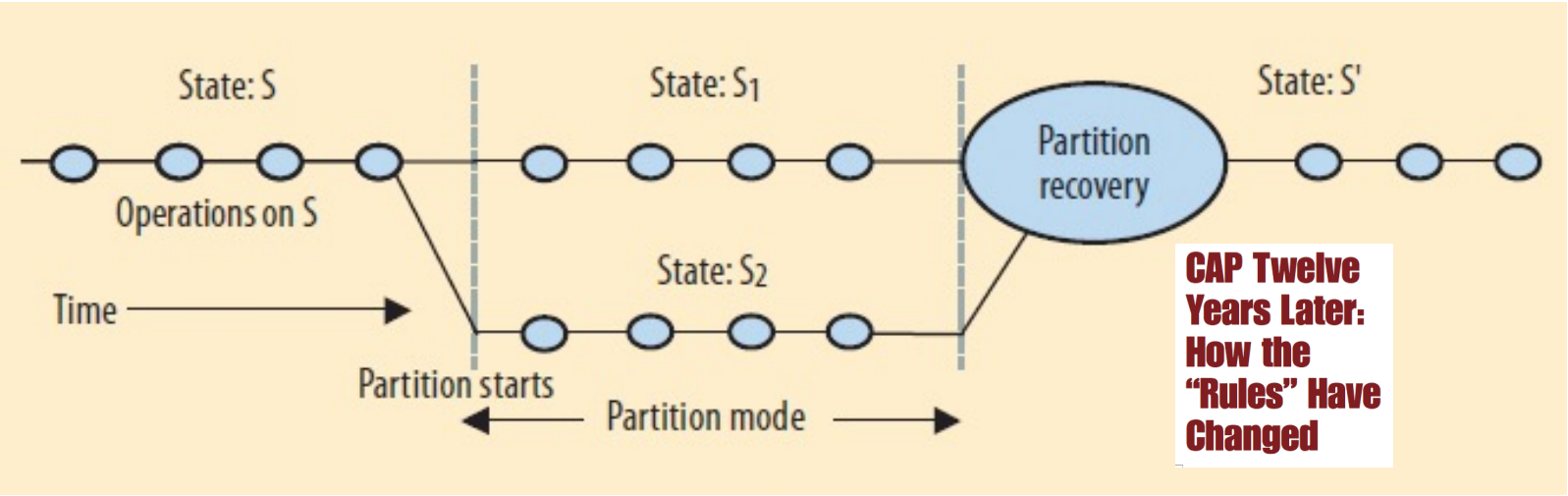
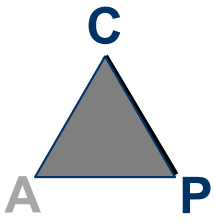


Figure 1. The state starts out consistent and remains so until a partition starts. To stay available, both sides enter **partition mode** and continue to execute operations, creating concurrent states S1 and S2, which are inconsistent. When the partition ends, the truth becomes clear and partition recovery starts. During recovery, the system merges **S1** and S2 into a consistent state S' and compensates for any mistakes made during the partition.

Image borrowed from Eric Brewer *CAP 12 Years Later (2012)*

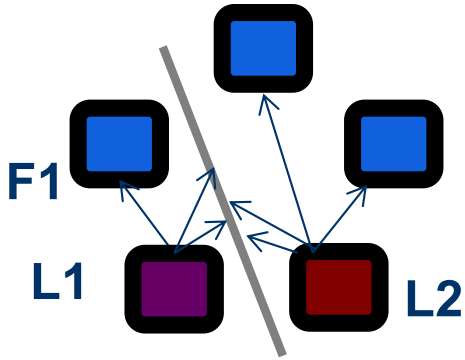
Consensus: Quorum rule



In order to serve and commit, a leader L1 must continually receive votes from a **majority**: f+1 of the full 2f+1 group (a **quorum**).

- **Safe** (C). In a partition scenario, at most one subgroup can serve requests.
- But not **live** (A). If a majority of replicas fail, it might be safe for the survivors to serve. But they **must not** because they cannot rule out a partition.

L1 and F1 do not have a quorum. So they must not serve clients → no **A**vailability if the majority had merely failed.

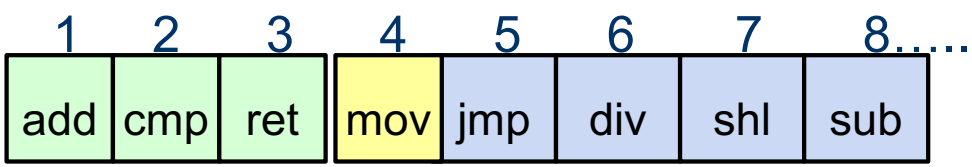


L2 can gain a quorum from the other followers, so its side of the partition can make progress.

At most one side of a partition makes progress: **disjoint majorities cannot exist**.

The operation log

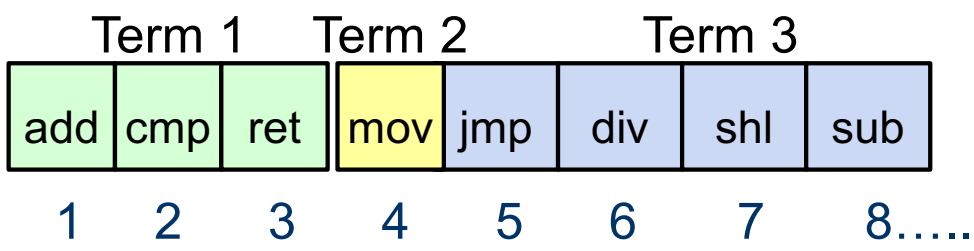
- Each replica has a local log of accepted actions (ops, commands).
- An operation log is a sequence of consecutively numbered entries.
  - Raft: log entry / log **index**. (Paxos: slots, VR: op-numbers)
  - Each entry has a corresponding action/op/command.
- Goals:
  - Agree on an action/op/command for each index (consensus).
  - Recognize when it is safe to **commit**: extend the committed log prefix.



Graphic from Raft authors

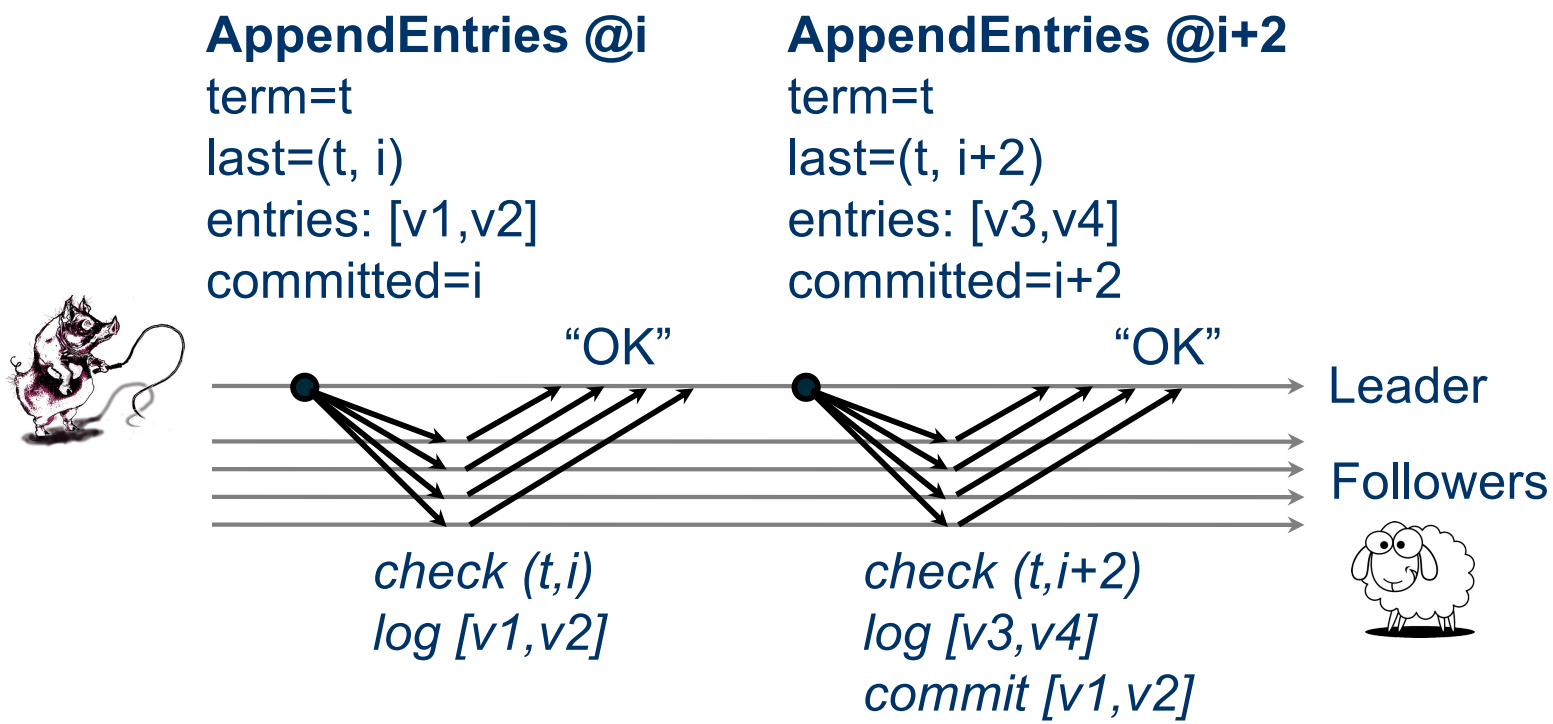
Terms/views

- The consensus protocol runs as a sequence of **terms**.
  - VR calls them **views**
- Terms are numbered: monotonically increasing.
  - Any replica can declare a new term on suspicion the leader has failed.
- Every message and log entry is tagged with a term#.
  - (term, index) pair is called a **viewstamp** in VR



Graphic from Raft authors

Raft in normal operation (stable term)

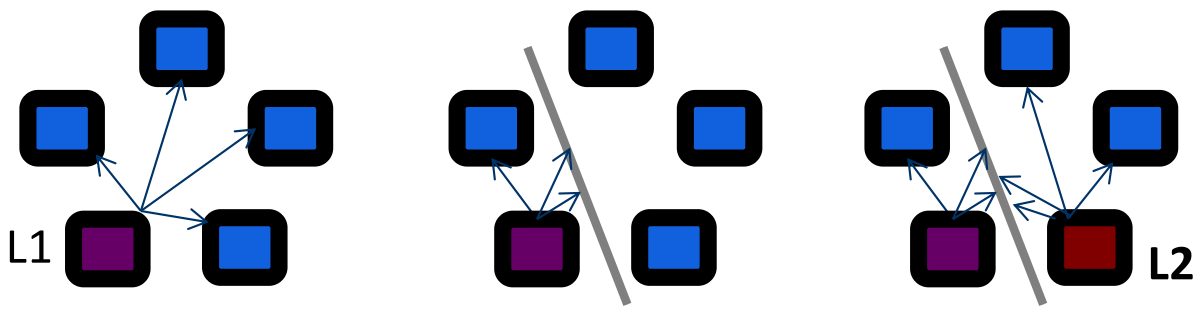


Followers accept what their leader tells them, even if it rewrites history. But first: check that leader's last viewstamp (term, index) is "in sync".

But partitions! A nasty scenario…

Consider a network partition: L1 survives. Split brain?

- How to make progress where we can do it safely?
- How to block progress where it is unsafe?



A happy group of 5 replicas.

Oops, partition.

Majority side elects L2; L1 struggles on.

Replica mode transitions

- Replicas keep their current role in a mode state variable.
- Haven't heard from your leader for "too long"? Timeout!
- Timeout: initiate leader election, declare new term++.
- Quorum → at most one leader can commit per view/term.

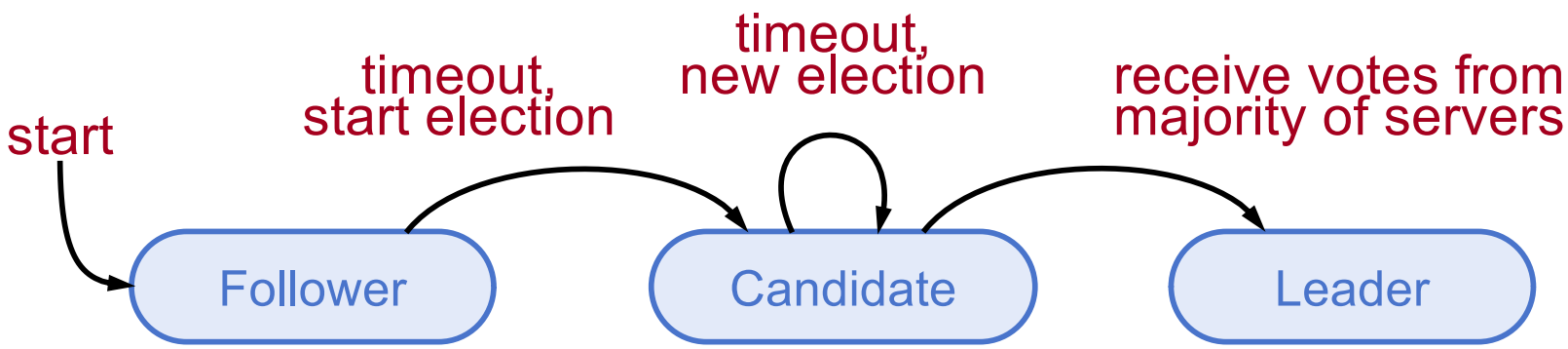
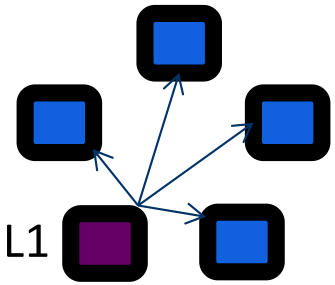


Image from authors of Raft Consensus Algorithm

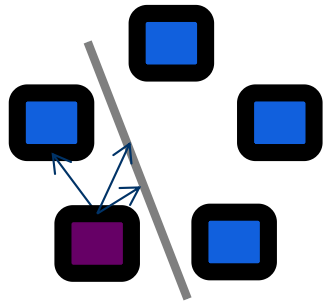
A nasty scenario continued: network heals

Partition heals → L1 and L2 both try to lead.

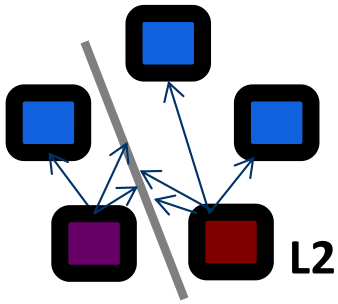
- How to establish which leader is "current"?
- How to reconcile/repair conflicting histories in the logs?
- How to ensure all committed entries survive?



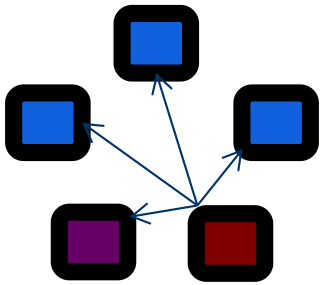
A happy group of 5 replicas.



Oops, partition.



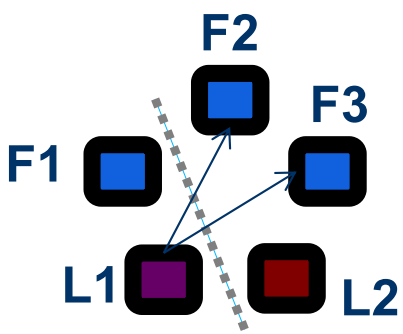
Majority side elects L2; L1 struggles on.



Partition heals. **Two leaders?**

Handling stale views/terms

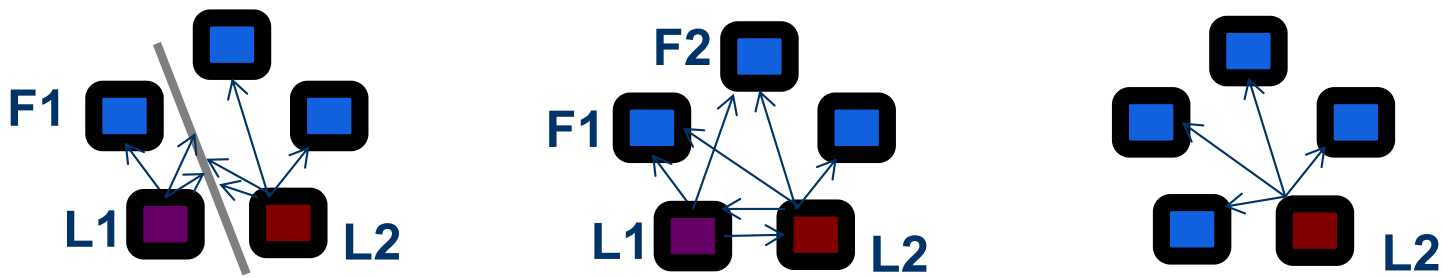
- Every message in the protocol carries the sender's term.
- If the receiver's term is higher, it ignores/rejects the message.
  - Stale participants can do no damage.
- If a participant learns of a term/view v higher than its own (stale!), it sets its term to v…
- (Yes, the term/view is an example of a **logical clock**.)



The net heals, and L1 reappears suddenly! But F2 and F3 are following a new leader L2 now. They say to L1: "I won't follow you."

Rejecting a leader from the past

- Replicas that hear L2 adopt L2's term—the latest and greatest.
- Those replicas ignore old leader L1: its term has expired—sad!
- L1 steps down when it learns of a term later than its own.



During the partition, L2 declares candidacy in a new term and wins a quorum. When the partition heals, F1 hears L2 is the new leader in town, and defects to L2. F2 ignores L1, whose term has expired. L1 learns of the new term and steps down.

Replica mode transitions

- Replicas transition state upon hearing of a new term.
- Even the followers defect to a new leader in town.

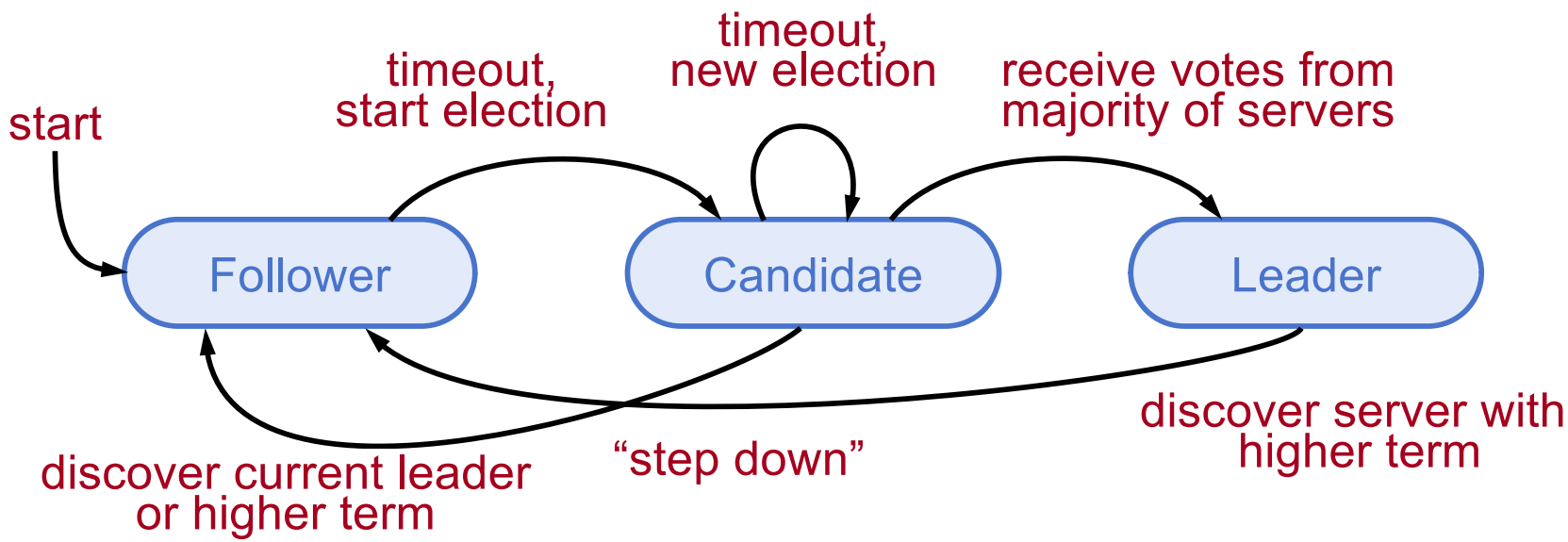
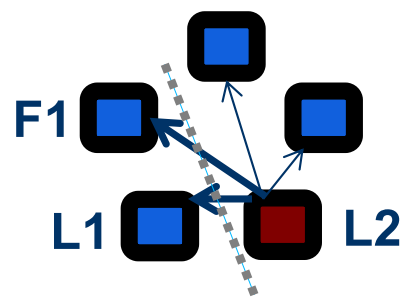


Image from authors of Raft Consensus Algorithm

Catching up

- After the network heals, L2 adopts new followers: F1 and L1.
- The new followers reject L2's append calls: "catch me up".
- L2 uses **repair-**appends to overwrite their log tails with its own.
- Visible to clients?

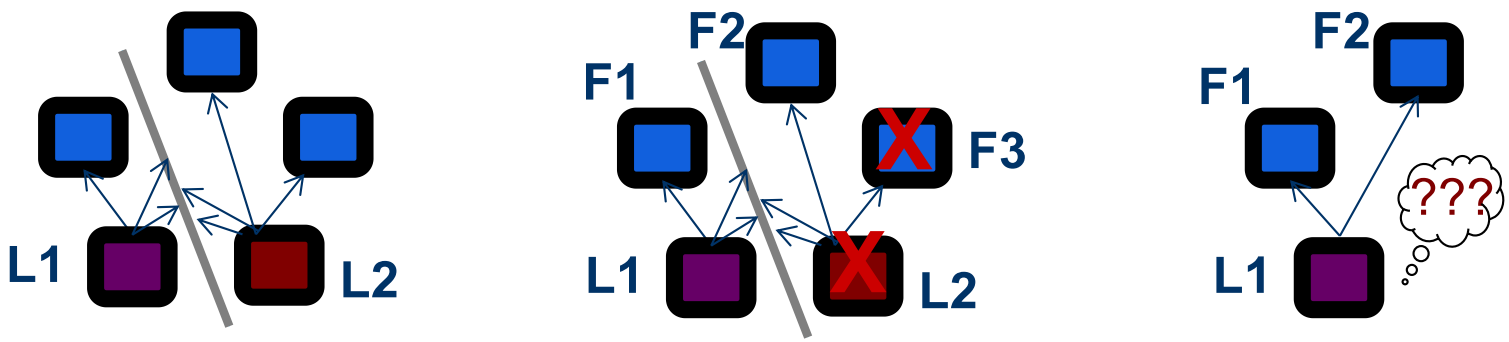
"Catch me up!"



The replicas regroup with L2 continuing as leader after the partition heals. L2 discovers that the nodes isolated on the minority side lag behind. L2 appends its log tail onto the new followers, starting from the common log prefix.

Nasty scenario: still nastier

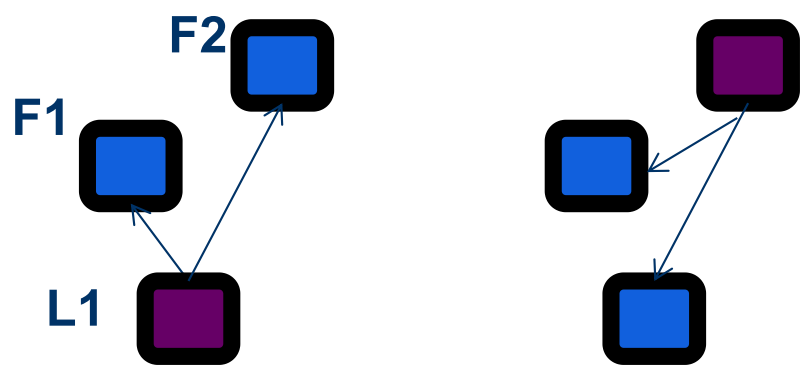
- **Suppose** as before: a bare majority commit new entries under L2.
- The minority writes new uncommitted entries under L1.
- The net heals and the minority rejoins, **but then L2 and a follower (F3) fail**!
- The survivors can make progress—if they can patch up their differences.
- Who knows the "right" committed history? How to propagate it?



Nasty scenario: still nastier

- Log repair considers the term at each log index (the **viewstamps**).
- As before: F2 rejects L1 and L1 steps down. (F2's term/viewstamp wins.)
- A new election is declared → new term with a new leader. **But who**?
- F2 has log entries from a higher term than the others. Higher terms/viewstamps dominate → the others accept F2's entries.
- **In Raft**, F2 declines to vote for a lagging candidate → becomes leader itself.

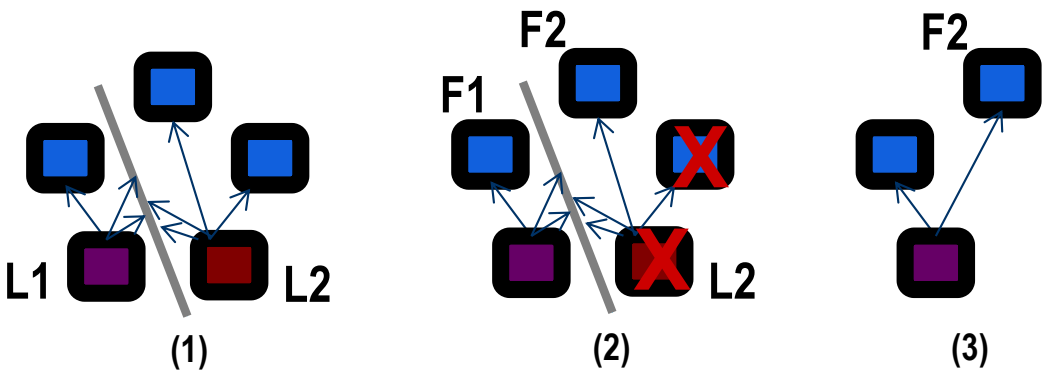
F2 rejects L1.
L1 steps down.
A new term begins.
An election starts.



Suppose F2 wins. (In Raft, it will). As leader, it pushes its log onto the others, rewriting their history.

Safety: committed entries survive into future views

- In our scenario, one replica (F2) survives to propagate commits into a new term.



- **Provable**: for **any** committed log entry *e*, **at least one** replica who knows of *e* survives into any future term (that has a new leader), always.
- Why? It requires a quorum to commit entries or elect a new leader, and any two quorums (quora?) **must** intersect.

Consensus design choices

Raft picks an up-to-date replica as leader

If log entries prepare/decide sequentially (as in Raft, VR, and Multi-Paxos) then a replica's viewstamp captures how current that replica is.

- Viewstamp is: (highest term, highest index for term)
- Within any voting majority, we can identify a replica R whose viewstamp is at least as up to date as all of the others.
- In VR, new leader is round robin but adopts the log of R in its entirety.
- In Multi-Paxos…well, go read it and see if you can figure out what it does.
- **In Raft, only such an "all caught up" R can win a vote for leader.**
  - A replica V votes for candidate C only if C is at least as up to date as V.
  - No need to transfer state to a new leader→one-way state transfer.

Summary

- Leader/master coordinates, dictates consensus
  - View the service as a deterministic state machine.
  - Master (also called "primary") dictates order of client operations.
  - → All non-faulty replicas reach the same state.
- Remaining problem: who is the leader?
  - Leader itself might fail or be isolated by a network partition.
  - Requires a scheme for "leader election" to choose a new leader.
  - Consensus is safe but not live: in the worst case (multiple repeated failures) it might not terminate.
  - But in practice Consensus gets the job done…if it can be done.

Raft: safety

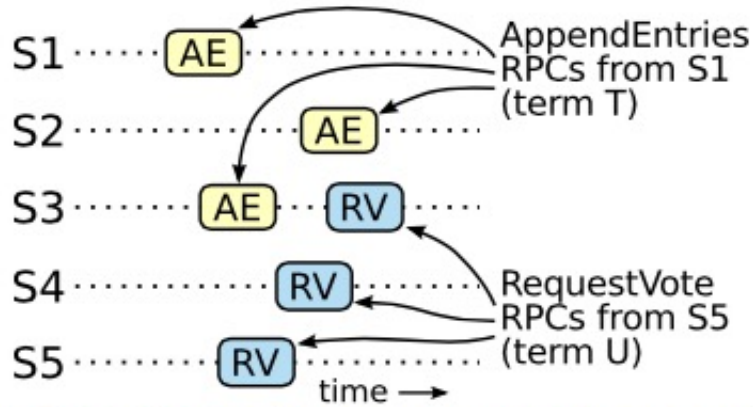


Figure 9: If S1 (leader for term T) commits a new log entry from its term, and S5 is elected leader for a later term U, then there must be at least one server (S3) that accepted the log entry and also voted for S5.

Graphic from Raft paper

Note: when a new leader L propagates log entries into a new term, L does not and cannot (always) know if they committed or not in an older term. But it does not matter: the set of surviving entries is guaranteed to include all committed entries, whether known or not. And once L propagates them to a new quorum, they are known to have committed.

The replicated log (Raft)

This Raft pic shows how the logs advance in a wavefront.

At any given time the replicas agree on a common log prefix.

This figure applies to other Consensus variants.

Index ←→ slot ←→ op-number
Term ←→ ballot ←→ view

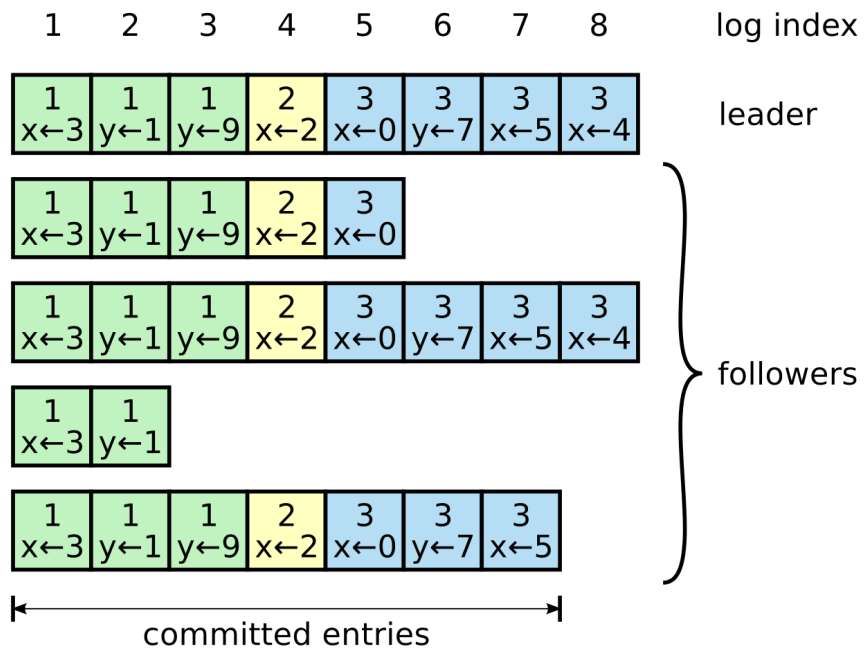


Figure 6: Logs are composed of entries, which are numbered sequentially. Each entry contains the term in which it was created (the number in each box) and a command for the state machine. An entry is considered *committed* if it is safe for that entry to be applied to state machines.

Graphic from Raft paper

Committed?
Who knows?

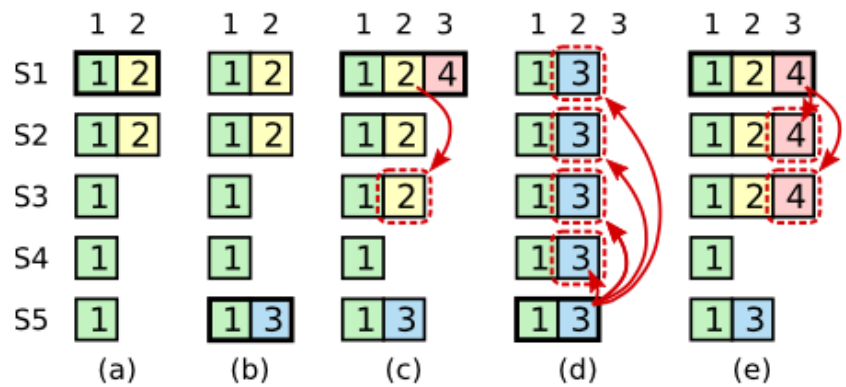


Figure 8: A time sequence showing why a leader cannot determine commitment using log entries from older terms. In (a) S1 is leader and partially replicates the log entry at index 2. In (b) S1 crashes; S5 is elected leader for term 3 with votes from S3, S4, and itself, and accepts a different entry at log index 2. In (c) S5 crashes; S1 restarts, is elected leader, and continues replication. At this point, the log entry from term 2 has been replicated on a majority of the servers, but it is not committed. If S1 crashes as in (d), S5 could be elected leader (with votes from S2, S3, and S4) and overwrite the entry with its own entry from term 3. However, if S1 replicates an entry from its current term on a majority of the servers before crashing, as in (e), then this entry is committed (S5 cannot win an election). At this point all preceding entries in the log are committed as well.

Graphic from Raft paper

Safety properties

Election Safety: at most one leader can be elected in a given term. §5.2

Leader Append-Only: a leader never overwrites or deletes entries in its log; it only appends new entries. §5.3

Log Matching: if two logs contain an entry with the same index and term, then the logs are identical in all entries up through the given index. §5.3

Leader Completeness: if a log entry is committed in a given term, then that entry will be present in the logs of the leaders for all higher-numbered terms. §5.4

State Machine Safety: if a server has applied a log entry at a given index to its state machine, no other server will ever apply a different log entry for the same index. §5.4.3

Figure 3: Raft guarantees that each of these properties is true at all times. The section numbers indicate where each property is discussed.

Consensus: stuff to know

- tolerate up to f failures; 2f+1 replicas → surviving f+1 is a quorum
- At most one viable leader at at any given time.
  - There might be multiple nodes in LEADER state, but at most one has a functioning quorum that can commit entries.
- Safety: the leader protects/preserves committed history.
  - Monotonic terms and quorum rules make it work, always.
- Election and commitment require a majority/quorum of the full configured group, not just of those present.
  - Quorum is key to preserving consistency/safety under partition.
- Not for scale. Adding replicas to an RSM group makes you do more work, but work doesn't get done any faster.

AppendEntries RPC

Invoked by leader to replicate log entries (§5.3); also used as heartbeat (§5.2).

Arguments:

term	leader's term
leaderId	so follower can redirect clients
prevLogIndex	index of log entry immediately preceding new ones
prevLogTerm	term of prevLogIndex entry
entries[]	log entries to store (empty for heartbeat; may send more than one for efficiency)
leaderCommit	leader's commitIndex

Results:

term	currentTerm, for leader to update itself
success	true if follower contained entry matching prevLogIndex and prevLogTerm

Receiver implementation:

1. Reply false if term < currentTerm (§5.1)
2. Reply false if log doesn't contain an entry at prevLogIndex whose term matches prevLogTerm (§5.3)
3. If an existing entry conflicts with a new one (same index but different terms), delete the existing entry and all that follow it (§5.3)
4. Append any new entries not already in the log
5. If leaderCommit > commitIndex, set commitIndex = min(leaderCommit, index of last new entry)

See Figure 2

Persistent state on all servers:
(Updated on stable storage before responding to RPCs)

currentTerm	latest term server has seen (initialized to 0 on first boot, increases monotonically)
votedFor	candidateId that received vote in current term (or null if none)

Volatile state on all servers:

commitIndex	index of highest log entry known to be committed (initialized to 0, increases monotonically)
lastApplied	index of highest log entry applied to state machine (initialized to 0, increases monotonically)

Volatile state on leaders:
(Reinitialized after election)

nextIndex[]	for each server, index of the next log entry to send to that server (initialized to leader last log index + 1)
matchIndex[]	for each server, index of highest log entry known to be replicated on server (initialized to 0, increases monotonically)

RequestVote RPC

Invoked by candidates to gather votes (§5.2).

Arguments:

term	candidate's term
candidateId	candidate requesting vote
lastLogIndex	index of candidate's last log entry (§5.4)
lastLogTerm	term of candidate's last log entry (§5.4)

Results:

term	currentTerm, for candidate to update itself
voteGranted	true means candidate received vote

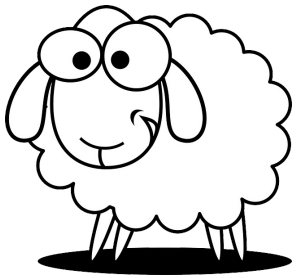
Receiver implementation:

1. Reply false if term < currentTerm (§5.1)
2. If votedFor is null or candidateId, and candidate's log is at least as up-to-date as receiver's log, grant vote (§5.2, §5.4)

The players

Leader/Primary

1. Become leader.
2. Rewrite history.
3. Dictate the future.
4. Stay leader forever.
5. If deposed goto step 1.

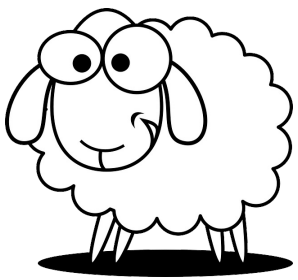


Acceptor/Replica/**Follower**

1. Adopt leader.
2. Tell it your history.
3. Accept whatever the leader says.
4. Write it all down.
5. If a new leader appears, goto step 1.

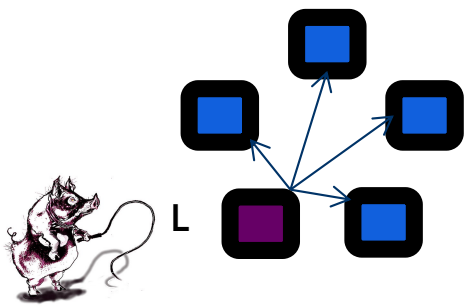
Consensus oversimplified

1. If leader falters try to take over.
2. Become leader by majority vote/assent.
3. Discover and affirm history.
4. Propose values for new log slots in order; notify others if majority accepts.

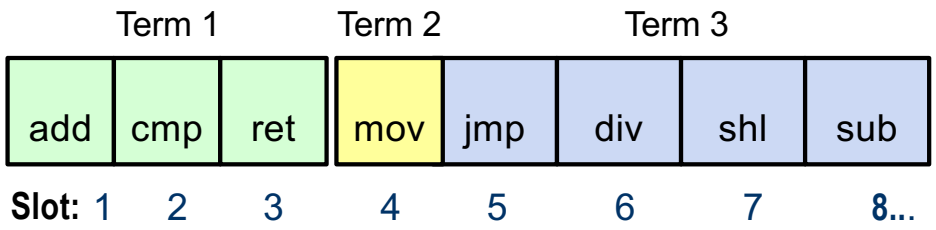


1. Adopt leader.
2. Tell it your history.
3. Accept whatever the leader says.
4. Write it all down.
5. If a new leader appears, goto step 1.

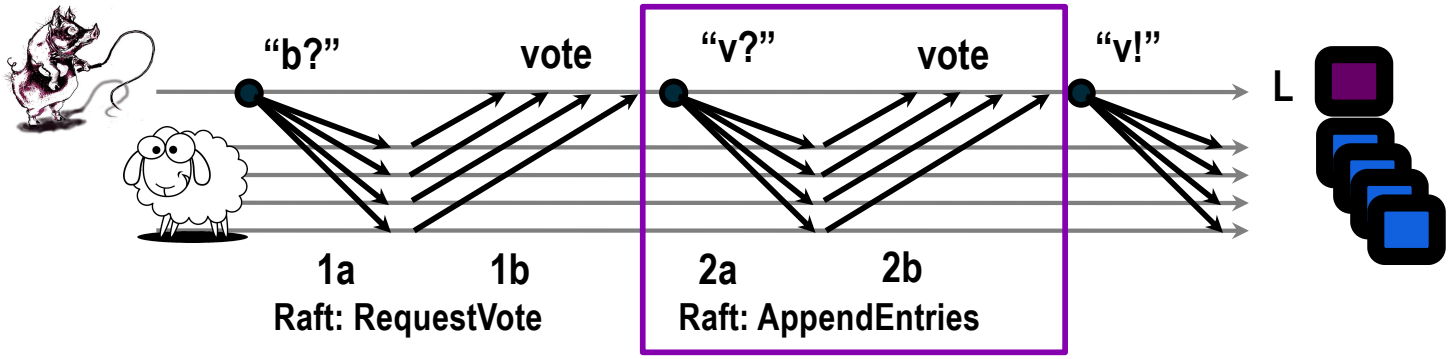
Consensus: review



A happy group of 5 replicas.



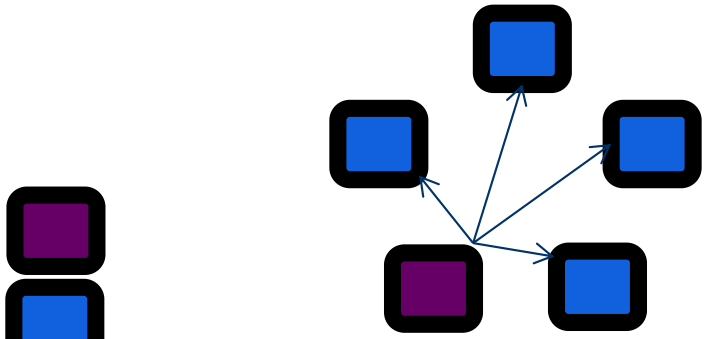
Each has a copy of a log of (ballot, value) pairs.
Term == ballot == view, index== slot



Notes

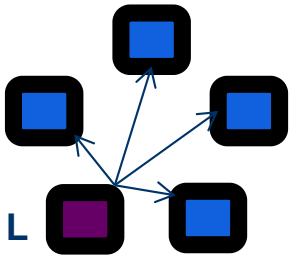
Replicating a service

- Replication → highly **available** services: "always up".
  - Run multiple copies (**replicas**).
  - Each has the same state.
  - They comprise a **replica group**.
  - One replica is **primary** or **leader**.
  - The others are **backup** replicas.
  - (Or standby or secondary or **follower**.)



- How to make it work for "any" service? ("reusable")

Notes

State Machine Replication (SMR/RSM)

- N copies (replicas) of a **deterministic** server.
- Replicas run the same app program P with the same API. It doesn't matter what P is, if it is deterministic.
  - P is a **state machine**: function {state, action}→state
- Each replica keeps its own copy of internal app state.
- Suppose they agree on a sequence (or log) of actions.
  - Ordered list of (requests, commands, ops, values). Linearizable.
- Then they converge.
  - Use any up-to-date replica
  - Or use leader/primary with fail-over
  - Designate stable leader